

Wiener-maximizing trees of diameter $n - 6$: the sharp threshold and all equality cases

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Abstract

We determine the maximum Wiener index and every maximizing isomorphism class among trees of order $n \geq 8$ and diameter $n - 6$. For every $n \geq 14$, the unique maximizer is the balanced double broom obtained by adding three and four leaves to the ends of a path on $n - 7$ vertices, and its Wiener index is $(n^3 - 55n + 222)/6$. The threshold is sharp: at $n = 13$ there are three maximizing classes. For $n = 8, \dots, 13$ the maxima are 49, 76, 117, 160, 220, 284, with respectively 1, 1, 1, 2, 1, 3 equality classes. A known peripheral-leaf lemma reduces the problem to 31 explicitly listed pairs of rooted end profiles. Their differences from the double broom are affine functions of the diameter, with exact validity domains. Eventual double-broom optimality is prior work; the contribution here is the sharp classification and its complete finite certificate.

1 Statement and relation to prior work

All trees are finite, simple, and undirected. Distances count edges, and the Wiener index counts each unordered pair once:

$$W(T) = \sum_{\{u,v\} \subseteq V(T), u \neq v} \text{dist}_T(u, v).$$

For $n \geq 8$, let $M_6(n)$ be the maximum of $W(T)$ over all trees of order n and diameter $d = n - 6$. Let B_n be obtained from a path on $n - 7$ vertices by attaching three new leaves to one end and four to the other. Thus $\text{diam}(B_n) = n - 6$. When $n = 8$ the two ends coincide and $B_8 = K_{1,7}$.

Theorem 1. *For every integer $n \geq 14$,*

$$M_6(n) = \frac{n^3 - 55n + 222}{6},$$

and equality holds precisely for $T \cong B_n$. For $8 \leq n \leq 13$, all maximizing isomorphism classes are given in Table 1. In particular, 14 is the least threshold above which B_n is always the unique maximizer.

Borovićanin, Božović, Glogić, Mesarić Štesl, Špacapan, and Zogić [1] prove that every leaf of a Wiener-maximizing tree is peripheral. Their Theorem 2.13 gives the balanced-double-broom characterization when

$$n \geq 1636, \quad n \leq d - 2 + 4 \left\lfloor \sqrt{\frac{d-1}{2}} \right\rfloor.$$

Table 1: Complete exceptional classification. The explicit profile construction $T_d(L, R)$ is defined in Section 2; the profiles are listed in Table 5.

n	$M_6(n)$	All maximizing classes
8	49	B_8
9	76	B_9
10	117	$T_4(P_{1,1}, P_{4,3})$
11	160	$T_5(P_{2,1}, P_{3,2}), T_5(P_{1,1}, P_{4,3})$
12	220	$T_6(P_{0,1}, P_{5,7})$
13	284	$B_{13}, T_7(P_{2,1}, P_{3,2}), T_7(P_{1,1}, P_{4,4})$

For $d = n - 6$ this includes every $n \geq 1636$. Their introduction also reports a result of Sun et al. whose stated bound specializes to $n \geq 166$. We have not inspected the original 2019 article and use neither that reported cutoff nor its proof as a dependency here.

Pandey and Ravi [2] prove the necessary condition that every vertex of an extremal tree lies on some diameter path (Theorem 3.8). Their Proposition 3.5 concerns caterpillars, and their discussion treats diameters $n - 4$ and $n - 5$ and suggests examining the next fixed deficits. The cited primary texts also discuss earlier solutions for diameters at most six. Accordingly, neither eventual stability nor the small values with $n \leq 12$ are claimed separately as new here. Our result is the complete deficit-six classification, including all equality cases and the least stable uniqueness threshold, with a 31-row certificate. No equivalent complete classification was located in the two inspected primary texts and the associated arXiv search recorded on September 9, 2026. This limited comparison does not establish worldwide priority.

2 Reduction to 31 profile pairs

A vertex is *peripheral* if its eccentricity equals the diameter. We include a proof of the following known reduction to make the argument self-contained and identify its exact scope.

Lemma 2 ([1, Lemma 2.1]). *Every leaf of a tree maximizing the Wiener index at fixed order and diameter d is peripheral.*

Proof. Suppose that a leaf x has $\text{ecc}_T(x) < d$. Follow its path to the first vertex y of degree at least three; such a vertex exists since T cannot be a path. Deleting the edge of this path incident to y separates a pendant path S of order $k \geq 1$ from a tree R of order N . The vertex y has at least two neighbors in R . Choose a neighbor z whose component in $R - y$ has minimum order b . Then $b \leq (N - 1)/2$.

Detach S from y and attach its same endpoint to z . A diameter path of T cannot enter S , since it would end at x . Hence R still contains a path of length d . Each old S – R distance is at most $\text{ecc}_T(x) \leq d - 1$ and changes by exactly one; distances within S and within R are unchanged. The resulting tree T' therefore has diameter exactly d . For every vertex of S , its b distances into the chosen component decrease by one and its other $N - b$ distances to R increase by one. Thus

$$W(T') - W(T) = k(N - 2b) \geq k > 0,$$

contradicting maximality. □

Lemma 3. *Fix a diameter path $v_0 v_1 \cdots v_d$ in a tree. If the nearest path vertex to x is v_i and $h = \text{dist}(x, v_i)$, then $\text{ecc}(x) = h + \max(i, d - i)$.*

Proof. For a vertex y with projection v_j and height k , its distances to the two endpoints imply $k \leq \min(j, d - j)$. The walk through the projections gives

$$\text{dist}(x, y) \leq h + |i - j| + k \leq h + \max(i, d - i).$$

Indeed, use $k \leq j$ when $j \leq i$, and $k \leq d - j$ otherwise. One of v_0, v_d attains the claimed upper bound. The walk argument also applies when the two projections coincide. $\square$

Fix a diameter path in an extremal tree of order $n = d + 6$. Exactly five vertices lie outside it. Each component outside the path has one edge to an internal path vertex v_i ; a second edge would create a cycle, while an attachment to an endpoint would extend the diameter. Root the component at its vertex adjacent to v_i . Its rooted leaves are leaves of T . By Lemmas 2 and 3, every such leaf has distance $r = \min(i, d - i)$ from v_i . Thus all rooted leaves have depth $r - 1$, and $r \leq 5$.

For clarity we encode a rooted tree by a pair of parentheses containing its child codes in lexicographic order; a singleton has code $()$. Table 2 lists every rooted tree of order at most five whose leaves have equal depth. Its *radius* is one plus that depth.

Table 2: The twelve possible rooted components. Radius is measured from the attachment vertex on the diameter path.

Component	Order	Radius	Rooted code
C_1	1	1	$()$
C_2	2	2	$((()))$
C_3	3	2	$((())())$
C_4	4	2	$((())())()$
C_5	5	2	$((())()())$
C_6	3	3	$((()))()$
C_7	5	3	$((())()())$
C_8	4	3	$((())())()$
C_9	5	3	$((())()())$
C_{10}	4	4	$((())())()$
C_{11}	5	4	$((())()())$
C_{12}	5	5	$((())()())$

The list is exhaustive by the following elementary count. Radius one forces a singleton. At radius two the root has one through four leaf children. At radius three every child subtree needs at least two vertices: there is either one child, itself a star with one through three leaves, or two children each having one leaf. At radius four the four-vertex chain may acquire only one extra terminal leaf; earlier branching would require at least two extra vertices to keep all depths equal. Radius five forces the five-vertex chain. Greater radius is impossible.

An *end profile* is a multiset of these components of total order $a \leq 5$. Its components retain their radii. The numbers of profiles for $a = 0, 1, \dots, 5$ are 1, 1, 2, 4, 8, 15; all are explicitly listed in Table 5. Completeness can also be checked by expanding through degree five

$$\frac{1}{(1 - z)(1 - z^2)(1 - z^3)^2(1 - z^4)^3(1 - z^5)^5}.$$

Write $P_{a,j}$ for the j th profile of order a in that table.

Given profiles L, R of orders a, b with $a + b = 5$, form $T_d(L, R)$ by starting with $v_0 \cdots v_d$ and attaching each component of L at v_r and each component of R at v_{d-r} , where r is its radius. Each attachment adds an edge to the component root. Impose

$$d \geq d_{\min} := 2r_{\max}, \tag{1}$$

where $r_{\max}$ is the largest radius in either profile. Midpoint attachments from different profiles remain distinct components attached to the same path vertex.

Condition (1) is exact. Under it, every component leaf has distance d to the opposite path endpoint. Leaves on one side with radii $r \leq s$ have distance at most $r + (s - r) + s = 2s \leq d$; the same bound holds within a component by walking through the attachment. Leaves on opposite sides have distance $r + (d - r - s) + s = d$. Thus the diameter is d and every leaf is peripheral. Conversely, a component of radius $r > d/2$ has a leaf at distance $2r > d$ from its designated path end, whenever its attachment is defined. Hence it cannot give a diameter- d tree.

Conversely, any tree of the required order whose leaves are peripheral has this form. Assign each off-path component to the nearer end of the chosen diameter path, assigning midpoint components to either end. Reflect the path to arrange $a \leq b$. The possible order pairs are $(0, 5), (1, 4), (2, 3)$, giving

$$1 \cdot 15 + 1 \cdot 8 + 2 \cdot 4 = 31$$

profile pairs. This is a cover of all possible maximizers at every $n \geq 8$. It need not be injective: a tree may have several diameter paths or several midpoint assignments. We retain all descriptions until the equality classification.

3 Exact comparison for every diameter

For an extra vertex x in a left profile L , let t_x be its depth from its path attachment and r_x that attachment's radius. Define

$$S_L = \sum_{x \in L} t_x, \quad I_L = \sum_{x \in L} r_x, \quad Q_L = \sum_{x \in L} r_x^2, \quad \delta_L = S_L - I_L,$$

and let E_L be the sum of distances over unordered pairs of extra vertices in L . These distances depend only on the profile. Define $S_R, I_R, Q_R, \delta_R, E_R$ similarly. Since $t_x \leq r_x$, $\delta_L, \delta_R \leq 0$. Table 5 gives all parameters.

The path contributes $d(d+1)(d+2)/6$. An extra vertex of depth t at radius r has total distance to the path

$$(d+1)t + \frac{r(r+1)}{2} + \frac{(d-r)(d-r+1)}{2} = \frac{d(d+1)}{2} + d(t-r) + t + r^2.$$

For a left-right pair x, y the distance is $d + (t_x - r_x) + (t_y - r_y)$, including coincident midpoint attachments. Set

$$F(d) = \frac{d(d+1)(d+2)}{6} + \frac{5d(d+1)}{2}.$$

Partitioning all unordered vertex pairs now proves the identity

$$\begin{aligned} W(T_d(L, R)) &= F(d) + d(ab + \delta_L + \delta_R) \\ &\quad + S_L + S_R + Q_L + Q_R + E_L + E_R + b\delta_L + a\delta_R. \end{aligned} \tag{2}$$

This is valid on the exact domain (1); no interpolation is involved.

The pair $(P_{2,1}, P_{3,1})$ consists of two and three radius-one singletons and gives B_{d+6} . Here $ab = 6$, both deltas vanish, $S_L + S_R = Q_L + Q_R = 5$, and $E_L + E_R = 8$. Consequently

$$W(B_n) = F(d) + 6d + 18 = \frac{n^3 - 55n + 222}{6}, \quad n = d + 6. \tag{3}$$

Subtracting (2) yields

$$\begin{aligned} W(B_n) - W(T_d(L, R)) &= Ad + B, \\ A &= 6 - ab - \delta_L - \delta_R, \\ B &= 18 - S_L - S_R - Q_L - Q_R - E_L - E_R - b\delta_L - a\delta_R. \end{aligned} \tag{4}$$

Since $ab \leq 6$ and both deltas are nonpositive, $A \geq 0$. Equality forces $(a, b) = (2, 3)$ and both deltas zero. Every component of radius at least two has its root at depth $1 < r$, so contributes a strictly negative delta. Thus $A = 0$ forces the double-broom pair, for which also $B = 0$.

Table 6 gives $d_{\min}, A, B$ for all 31 pairs by direct substitution in (4). In every nonbroom row, $A > 0$ and

$$A \max(8, d_{\min}) + B > 0;$$

the smallest value is one. Therefore every valid nonbroom row is strictly inferior to B_n for every $d \geq 8$. Lemma 2, the complete cover, and the admissibility of B_n prove the stable assertion of Theorem 1. The domains matter: the radius-five row has gap $16d - 142$ and starts at $d = 10$. Its negative formal value at $d = 8$ is outside its domain; its first valid gap is 18.

4 Small orders and equality identifications

For each $d = 2, \dots, 7$, retain the rows with $d_{\min} \leq d$ and minimize $Ad + B$. Table 3 records every minimizing pair. Together with (3), it gives precisely the six exceptional values in Table 1.

Table 3: All minimizing descriptions at small orders. The second column is $W(B_n) - M_6(n)$. Distinct descriptions may represent the same tree.

n	Gap	All minimizing pairs (L, R)
8	0	$(P_{0,1}, P_{5,1}), (P_{1,1}, P_{4,1}), (P_{2,1}, P_{3,1})$
9	0	$(P_{2,1}, P_{3,1})$
10	-5	$(P_{1,1}, P_{4,3})$
11	-2	$(P_{2,1}, P_{3,2}), (P_{2,1}, P_{3,3}), (P_{1,1}, P_{4,3})$
12	-5	$(P_{0,1}, P_{5,7}), (P_{1,1}, P_{4,4}), (P_{1,1}, P_{4,7})$
13	0	$(P_{2,1}, P_{3,1}), (P_{2,1}, P_{3,2}), (P_{2,1}, P_{3,3}), (P_{1,1}, P_{4,4}), (P_{1,1}, P_{4,7})$

At $n = 8$ all descriptions give the star. At $n = 11$, $(P_{2,1}, P_{3,2})$ and $(P_{2,1}, P_{3,3})$ describe the same tree. At $n = 12$ all three minimizing descriptions are isomorphic. At $n = 13$, the descriptions with right profiles $P_{4,4}$ and $P_{4,7}$ are isomorphic, as are those with $P_{3,2}$ and $P_{3,3}$. These identifications give Table 1.

Here is an exact way to check both these identifications and the absence of further ones. The rooted code defined above is a complete invariant: induction on order shows that equal codes give matching multisets of rooted child subtrees, hence a root-preserving isomorphism. The lexicographically least rooted code over all choices of root is therefore a complete invariant of an unrooted tree. Computing it directly from the displayed component constructions gives Table 4. Every minimizing pair in the same group just described has the same code. The two codes at $n = 11$ and the three at $n = 13$ are different. This proves the equality counts, and the three classes at $n = 13$ show that the threshold 14 cannot be decreased.

Table 4: Distinct canonical codes of all exceptional maximizers. The class number is only a local enumeration within each order.

n	Class	Least rooted code
8	1	((()()()()()))
9	1	((()()()()()()))
10	1	((()()()()()()()))
11	1	((()()()()()()()()))
11	2	((()()()()()()()()))
12	1	((()()()()()()()()()))
13	1	((()()()()()()()()()()))
13	2	((()()()()()()()()()()))
13	3	((()()()()()()()()()()()))

5 Reproducibility and assistance

The proof uses the exhaustive component list, the exact distance identity, and the two finite parameter tables. The supplementary programs and certificates provide additional checks. The profile generator constructs the rooted trees recursively, obtains all 31 pairs, and checks 182 instances by both distance sums and the edge-cut formula

$$W(T) = \sum_{e \in E(T)} s_e(n - s_e),$$

where deleting e leaves a component of order s_e . This identity follows by counting which unordered vertex pairs use each edge.

A separate exhaustive construction attaches every rooted forest of total order five to a diameter path and filters by its actual diameter, without using the peripheral-leaf lemma. For $8 \leq n \leq 17$ it checks 18,788 markings, 13,741 valid markings, and 6,568 distinct eligible trees, and agrees with the classification. For $8 \leq n \leq 15$ it also agrees with an enumeration by successive leaf attachment. The forest computation checks the strict exchange on 6,431 nonperipheral classes. Its general tree utilities are shared with the profile implementation, so these are different enumeration constructions, not wholly independent programs.

An additional verifier imports none of those programs. It enumerates small rooted components through Prüfer trees, derives every Wiener polynomial from edge cuts by exact symbolic summation, and checks 304 direct integer instances and all 16 explicit isomorphism maps for the exceptional minimizing descriptions. All arithmetic is integral or exact symbolic arithmetic; no finite range is extrapolated to prove the infinite assertion. Instructions for running copies in a separate output directory accompany the supplementary files.

As adjacent methodology, Zeng [3] combines structural tree arguments with exact finite certification in the different setting of edge-graceful labellings. That reference is not used as a theorem about Wiener indices.

OpenAI Codex was used for proof development, implementation, manuscript preparation, and a separate mathematical checking pass. That pass and its separately written verifier are additional reproducibility checks, not external peer review or proof-assistant formalization. The argument presented here, including the equality classification, is fully within the fixed diameter- $n - 6$ problem; no claim about other deficits or worldwide priority is made.

References

- [1] B. Borovićanin, D. Božović, E. Glogić, D. Mesarič Štesl, S. Špacapan, and E. Zogić, *New results on Wiener index of trees with a given diameter*, arXiv:2507.17885v2 (2025). <https://arxiv.org/abs/2507.17885v2>.
- [2] D. Pandey and P. S. Ravi, *On a conjecture of DeLaViña and Waller*, arXiv:2605.24855v1 (2026). <https://arxiv.org/abs/2605.24855v1>.
- [3] Zijian Zeng, *Odd-Order Trees with Exactly Three Vertices of Degree Two Are Edge-Graceful*, preprint (2026). <https://doi.org/10.2139/ssrn.7387319>.

A The complete profile parameters

The notation C_i^k means k distinct copies of component C_i , not a graph power. Radii and rooted codes are given in Table 2. The first subscript of $P_{a,j}$ gives its total number of extra vertices. The quantities S, I, Q, E are defined in Section 3.

Table 5: All end profiles of total order at most five and their parameters.

Profile	Component multiset	S	I	Q	E
$P_{0,1}$	$\emptyset$	0	0	0	0
$P_{1,1}$	C_1	1	1	1	0
$P_{2,1}$	C_1^2	2	2	2	2
$P_{2,2}$	C_2	3	4	8	1
$P_{3,1}$	C_1^3	3	3	3	6
$P_{3,2}$	$C_1 C_2$	4	5	9	8
$P_{3,3}$	C_3	5	6	12	4
$P_{3,4}$	C_6	6	9	27	4
$P_{4,1}$	C_1^4	4	4	4	12
$P_{4,2}$	$C_1^2 C_2$	5	6	10	17
$P_{4,3}$	$C_1 C_3$	6	7	13	15
$P_{4,4}$	$C_1 C_6$	7	10	28	19
$P_{4,5}$	C_2^2	6	8	16	14
$P_{4,6}$	C_4	7	8	16	9
$P_{4,7}$	C_8	9	12	36	9
$P_{4,8}$	C_{10}	10	16	64	10
$P_{5,1}$	C_1^5	5	5	5	20
$P_{5,2}$	$C_1^3 C_2$	6	7	11	28
$P_{5,3}$	$C_1^2 C_3$	7	8	14	28
$P_{5,4}$	$C_1^2 C_6$	8	11	29	36
$P_{5,5}$	$C_1 C_2^2$	7	9	17	28
$P_{5,6}$	$C_1 C_4$	8	9	17	24
$P_{5,7}$	$C_1 C_8$	10	13	37	30
$P_{5,8}$	$C_1 C_{10}$	11	17	65	36
$P_{5,9}$	$C_2 C_3$	8	10	20	24
$P_{5,10}$	$C_2 C_6$	9	13	35	32
$P_{5,11}$	C_5	9	10	20	16
$P_{5,12}$	C_7	11	15	45	20
$P_{5,13}$	C_9	12	15	45	16
$P_{5,14}$	C_{11}	14	20	80	18
$P_{5,15}$	C_{12}	15	25	125	20

B The complete affine comparison

For every row, the tree is admissible exactly for $d \geq d_{\min}$, and $W(B_{d+6}) - W(T_d(L, R)) = Ad + B$. Let $D = \max(8, d_{\min})$. The last column gives the first valid gap in the stable range.

Table 6: All 31 profile pairs. The sole zero-slope row is the balanced double broom.

L	R	$d_{\min}$	A	B	$AD + B$
$P_{0,1}$	$P_{5,1}$	2	6	-12	36
$P_{0,1}$	$P_{5,2}$	4	7	-27	29
$P_{0,1}$	$P_{5,3}$	4	7	-31	25
$P_{0,1}$	$P_{5,4}$	6	9	-55	17
$P_{0,1}$	$P_{5,5}$	4	8	-34	30
$P_{0,1}$	$P_{5,6}$	4	7	-31	25
$P_{0,1}$	$P_{5,7}$	6	9	-59	13
$P_{0,1}$	$P_{5,8}$	8	12	-94	2
$P_{0,1}$	$P_{5,9}$	4	8	-34	30
$P_{0,1}$	$P_{5,10}$	6	10	-58	22
$P_{0,1}$	$P_{5,11}$	4	7	-27	29
$P_{0,1}$	$P_{5,12}$	6	10	-58	22
$P_{0,1}$	$P_{5,13}$	6	9	-55	17
$P_{0,1}$	$P_{5,14}$	8	12	-94	2
$P_{0,1}$	$P_{5,15}$	10	16	-142	18
$P_{1,1}$	$P_{4,1}$	2	2	-4	12
$P_{1,1}$	$P_{4,2}$	4	3	-15	9
$P_{1,1}$	$P_{4,3}$	4	3	-17	7
$P_{1,1}$	$P_{4,4}$	6	5	-35	5
$P_{1,1}$	$P_{4,5}$	4	4	-18	14
$P_{1,1}$	$P_{4,6}$	4	3	-15	9
$P_{1,1}$	$P_{4,7}$	6	5	-35	5
$P_{1,1}$	$P_{4,8}$	8	8	-62	2
$P_{2,1}$	$P_{3,1}$	2	0	0	0
$P_{2,1}$	$P_{3,2}$	4	1	-7	1
$P_{2,1}$	$P_{3,3}$	4	1	-7	1
$P_{2,1}$	$P_{3,4}$	6	3	-19	5
$P_{2,2}$	$P_{3,1}$	4	1	-3	5
$P_{2,2}$	$P_{3,2}$	4	2	-10	6
$P_{2,2}$	$P_{3,3}$	4	2	-10	6
$P_{2,2}$	$P_{3,4}$	6	4	-22	10